# THEORY AND APPLICATIONS OF MATRICES

A project submitted to **Madurai Kamaraj University, Madurai**

in partial fulfillment of the requirements for the award of the

Degree of

**BACHELOR OF SCIENCE**

**IN MATHEMATICS**

*by*


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**NOVEMBER - 2026**

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## CERTIFICATE

This is to certify that this project entitled, **"THEORY AND APPLICATIONS OF MATRICES"** submitted by

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for the Degree of the **BACHELOR OF SCIENCE IN MATHEMATICS** to Madurai Kamaraj University is based on the results of the studies carried out by the student under my guidance and supervision. It is further certified that all principles of academic honesty and integrity as advocated by Madurai Kamaraj University are strictly adhered to.

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## ACKNOWLEDGEMENT

We would like to thank God for having blessed us with courage and confidence to complete this research work successfully.

The successful completion of this undergraduate project, **"THEORY AND APPLICATIONS OF MATRICES"**, would not have been possible without the support, guidance, and encouragement of many individuals. It is with profound gratitude that we would like to acknowledge their contributions.

First and foremost, we wish to express our sincere gratitude to our project supervisor, **Dr. AR. MURUGAN**, Guest Lecturer , Department of Mathematics, Government Arts and Science College, Vedasandur, Dindigul-624 702. His unwavering guidance, insightful feedback, and immense patience were instrumental throughout this endeavour. His expertise steered us through challenges and His encouragement kept us motivated from the initial stages of research to the final draft.

We are also deeply thankful to the Principal, Head of the Department and staff members of Department of Mathematics, Government Arts and Science College, Vedasandur, Dindigul-624 702. The knowledge and foundation skills we gained from their courses provided the essential groundwork upon which this project was built.

Our heartfelt thanks go to Mr. M. MANIKANDAN, III B.Sc., Department of Mathematics, Government Arts and Science College, Vedasandur, Dindigul-624 702, our friends for their support. The stimulating discussions, shared resources, and moral support, especially during demanding phases, were invaluable. A special note of thanks is due to our class members for creating a collaborative and inspiring learning environment.

Finally, we owe our deepest debt of gratitude to our parents and our siblings for their unconditional love, unwavering belief in us, and endless sacrifices. Their constant encouragement has been our anchor and the driving force behind all our pursuits.

To all who have supported us directly or indirectly, please accept this as a token of our deepest appreciation. Thank you.

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# Chapter 1

# INTRODUCTION

## 1.1 Introduction to Matrices

Mathematics provides several methods for representing quantities and relationships. When a problem contains many related numbers, writing each number separately may become difficult. Matrices provide an organised way to store and manipulate such information.

A matrix is a rectangular arrangement of elements in horizontal rows and vertical columns. Its elements may be real numbers, complex numbers or other mathematical objects, depending on the context.

Matrices are widely used in linear algebra, differential equations, numerical analysis, physics, economics, computer science and engineering.

For example, the marks of three students in two subjects can be represented as

$$A = \begin{bmatrix} 80 & 75 \\ 65 & 90 \\ 72 & 85 \end{bmatrix}.$$

Each row represents a student, and each column represents a subject. Thus, a matrix gives a compact representation of the data.

## 1.2 Historical Background

The development of matrix methods is closely connected with the study of systems of linear equations. Determinants were investigated by mathematicians such as Seki Takakazu and Gottfried Wilhelm Leibniz. Later, Gabriel Cramer developed a determinant-based method for solving systems of equations.

Arthur Cayley made important contributions to the theory of matrices during the nineteenth century. Matrix algebra subsequently became an essential part of modern mathematics.

Today, matrices are used in scientific computing, computer graphics, artificial intelligence, statistics, engineering and many other fields.

## 1.3 Objectives of the Project

The objectives of this project are:

1. To understand the definition and notation of matrices.
2. To classify matrices according to their properties.
3. To study the basic operations on matrices.
4. To understand determinants and inverse matrices.
5. To learn methods for solving linear equations.
6. To study rank, eigenvalues and eigenvectors.
7. To examine important practical applications.

## 1.4 Scope of the Project

This project focuses on matrices with real-number entries and the basic concepts of finite-dimensional linear algebra. It introduces theoretical results and demonstrates their use through worked examples.

The project also provides an introduction to applications in data analysis, transformations, networks and mathematical modelling.

# Chapter 2

# FUNDAMENTALS OF MATRICES

## 2.1 Definition and Notation

A matrix is a rectangular array of elements arranged in rows and columns.

A matrix with $m$ rows and $n$ columns is written as

$$A = [a_{ij}]_{m \times n},$$

where $a_{ij}$ denotes the element in the $i$-th row and $j$-th column.
In expanded form,

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$$

## 2.2 Order of a Matrix

The order of a matrix is the number of rows followed by the number of columns.
For example,

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

has two rows and three columns. Therefore, its order is $2 \times 3$.
Similarly,

$$B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

has order $2 \times 2$.

## 2.3 Elements of a Matrix

Consider

$$A = \begin{bmatrix} 2 & 4 & 6 \\ 1 & 3 & 5 \\ 7 & 8 & 9 \end{bmatrix}.$$

The element in the second row and third column is $a_{23} = 5$. The element $a_{31} = 7$ lies in the third row and first column.
The first subscript indicates the row, and the second subscript indicates the column.

## 2.4 Types of Matrices

### 2.4.1 Row Matrix

A matrix containing only one row is called a row matrix.

$$A = \begin{bmatrix} 2 & 4 & 6 & 8 \end{bmatrix}.$$

Its order is $1 \times 4$.

### 2.4.2 Column Matrix

A matrix containing only one column is called a column matrix.

$$B = \begin{bmatrix} 3 \\ 5 \\ 7 \end{bmatrix}.$$

Its order is $3 \times 1$.

### 2.4.3 Rectangular Matrix

A matrix with different numbers of rows and columns is called a rectangular matrix.

$$C = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}.$$

### 2.4.4 Square Matrix

A matrix with the same number of rows and columns is called a square matrix.

$$D = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$

Its order is $2 \times 2$.

### 2.4.5 Zero Matrix

A matrix in which every element is zero is called a zero matrix.

$$O = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

### 2.4.6 Diagonal Matrix

A square matrix whose non-diagonal elements are zero is called a diagonal matrix.

$$D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 7 \end{bmatrix}.$$

### 2.4.7 Scalar Matrix

A diagonal matrix whose diagonal elements are equal is called a scalar matrix.

$$S = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}.$$

### 2.4.8 Identity Matrix

A square matrix with ones on its main diagonal and zeros elsewhere is called an identity matrix.

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

For every compatible matrix $A$,

$$AI = A, \qquad IA = A.$$

### 2.4.9 Upper and Lower Triangular Matrices

An upper triangular matrix has zero entries below the main diagonal.

$$U = \begin{bmatrix} 2 & 3 & 4 \\ 0 & 5 & 6 \\ 0 & 0 & 7 \end{bmatrix}.$$

A lower triangular matrix has zero entries above the main diagonal.

$$L = \begin{bmatrix} 2 & 0 & 0 \\ 3 & 5 & 0 \\ 4 & 6 & 7 \end{bmatrix}.$$

### 2.4.10 Symmetric Matrix

A square matrix $A$ is symmetric if

$$A^T = A.$$

For example,

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}.$$

### 2.4.11 Skew-Symmetric Matrix

A square matrix $A$ is skew-symmetric if

$$A^T = -A.$$

For example,

$$A = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}.$$

Every diagonal element of a skew-symmetric matrix over the real numbers is zero.

## 2.5 Equality of Matrices

Two matrices are equal if they have the same order and their corresponding elements are equal.

If

$$A = \begin{bmatrix} x & 2 \\ 3 & y \end{bmatrix}, \qquad B = \begin{bmatrix} 4 & 2 \\ 3 & 7 \end{bmatrix},$$

and $A = B$, then

$$x = 4, \qquad y = 7.$$

## 2.6 Transpose of a Matrix

The transpose of a matrix is obtained by interchanging its rows and columns.

If

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix},$$

then

$$A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}.$$

Important properties include

$$(A^T)^T = A,$$

$$(A + B)^T = A^T + B^T,$$

and

$$(AB)^T = B^T A^T.$$

The last identity reverses the order of the factors.

# Chapter 3

# ALGEBRA OF MATRICES

## 3.1 Addition of Matrices

Two matrices can be added only if they have the same order. Their corresponding elements are added.

Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}.$$

Then

$$A + B = \begin{bmatrix} 1+5 & 2+6 \\ 3+7 & 4+8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}.$$

## 3.2 Properties of Matrix Addition

For matrices of the same order,

$$A + B = B + A$$

and

$$(A + B) + C = A + (B + C).$$

The zero matrix satisfies

$$A + O = A.$$

Every matrix $A$ has an additive inverse $-A$, such that

$$A + (-A) = O.$$

## 3.3 Subtraction of Matrices

Subtraction is defined by adding the negative of a matrix.

Using the matrices above,

$$B - A = \begin{bmatrix} 5-1 & 6-2 \\ 7-3 & 8-4 \end{bmatrix} = \begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix}.$$

In general, $A - B$ need not equal $B - A$.

## 3.4 Scalar Multiplication

Multiplying every element of a matrix by a scalar produces a scalar multiple.

If

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix},$$

then

$$3A = \begin{bmatrix} 3 & 6 \\ 9 & 12 \end{bmatrix}.$$

The distributive properties are

$$k(A + B) = kA + kB$$

and

$$(k + \ell)A = kA + \ell A.$$

## 3.5 Matrix Multiplication

Matrix multiplication is defined when the number of columns of the first matrix equals the number of rows of the second matrix.

If $A$ has order $m \times n$ and $B$ has order $n \times p$, then $AB$ has order $m \times p$.

The element in row $i$, column $j$ of the product is

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}.$$

### 3.5.1 Worked Example

Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix}.$$

Then

$$AB = \begin{bmatrix} 1(2) + 2(1) & 1(0) + 2(5) \\ 3(2) + 4(1) & 3(0) + 4(5) \end{bmatrix}.$$

Therefore,

$$AB = \begin{bmatrix} 4 & 10 \\ 10 & 20 \end{bmatrix}.$$

## 3.6 Non-Commutativity of Multiplication

Matrix multiplication is generally not commutative.

For the same matrices,

$$BA = \begin{bmatrix} 2 & 4 \\ 16 & 22 \end{bmatrix}.$$

Thus,

$$AB \neq BA.$$

# Chapter 4

# DETERMINANTS

## 4.1 Introduction

A determinant is a scalar associated with a square matrix. It helps determine whether a matrix is invertible and provides methods for solving systems of equations.

The determinant of $A$ is written as $\det(A)$ or $|A|$.

## 4.2 Determinant of a Matrix of Order Two

For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

the determinant is

$$\det(A) = ad - bc.$$

### 4.2.1 Worked Example

Find the determinant of

$$A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}.$$

Using the formula,

$$\det(A) = 3(4) - 2(1) = 12 - 2 = 10.$$

Therefore, $\det(A) = 10$.

## 4.3 Determinant of a Matrix of Order Three

For a matrix of order three,

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix},$$

expansion along the first row gives

$$\begin{aligned} \det(A) = a(ei - fh) \\ - b(di - fg) \\ + c(dh - eg). \end{aligned}$$

### 4.3.1 Worked Example

Evaluate

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix}.$$

Expanding along the first row,

$$\begin{aligned} \det(A) = 1(1 \cdot 0 - 4 \cdot 6) \\ - 2(0 \cdot 0 - 4 \cdot 5) \\ + 3(0 \cdot 6 - 1 \cdot 5). \end{aligned}$$

Hence,

$$\det(A) = -24 + 40 - 15 = 1.$$

Therefore, $\det(A) = 1$.

## 4.4 Minors and Cofactors

The minor $M_{ij}$ is the determinant obtained by deleting row $i$ and column $j$.

The cofactor is defined by

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

For a matrix of order three, the cofactor signs are

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}.$$

Cofactors are used to calculate determinants and inverse matrices.

## 4.5 Expansion Along a Row or Column

The determinant can be expanded along any row or column.

Expansion along row $i$ gives

$$\det(A) = \sum_{j=1}^{n} a_{ij} C_{ij}.$$

Expansion along column $j$ gives

$$\det(A) = \sum_{i=1}^{n} a_{ij} C_{ij}.$$

These formulas are particularly useful when a row or column contains several zeros.

## 4.6 Properties of Determinants

Important properties include:

1. Interchanging two rows changes the sign of the determinant.
2. If two rows are identical, the determinant is zero.
3. Multiplying one row by $k$ multiplies the determinant by $k$.
4. Adding a multiple of one row to another row does not change the determinant.

# Chapter 5

# INVERSE OF A MATRIX

## 5.1 Definition

A square matrix $A$ is invertible if there exists a matrix $A^{-1}$ satisfying

$$AA^{-1} = A^{-1}A = I.$$

The matrix $A^{-1}$ is called the inverse of $A$.
A square matrix is invertible if and only if its determinant is non-zero.

## 5.2 Inverse of a Matrix of Order Two

For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

if $ad - bc \neq 0$, then

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

### 5.2.1 Derivation

Let

$$B = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Multiplication gives

$$AB = \begin{bmatrix} ad - bc & -ab + ab \\ cd - dc & ad - bc \end{bmatrix}.$$

Therefore,

$$AB = (ad - bc)I.$$

Dividing by the non-zero scalar $ad - bc$, we obtain

$$A\left(\frac{B}{ad - bc}\right) = I.$$

Thus the inverse formula follows.

## 5.3 Adjoint Method

The adjugate of a square matrix is the transpose of its cofactor matrix.

For an invertible matrix,

$$\boxed{A^{-1} = \frac{\operatorname{adj}(A)}{\det(A)}}.$$

This formula is valid when $\det(A) \neq 0$.

For large matrices, Gaussian elimination is often more efficient than explicitly computing all cofactors.

## 5.4 Properties of Inverse Matrices

For invertible matrices of compatible orders,

$$(AB)^{-1} = B^{-1}A^{-1},$$

$$(A^T)^{-1} = (A^{-1})^T,$$

$$(A^{-1})^{-1} = A,$$

and

$$\det(A^{-1}) = \frac{1}{\det(A)}.$$

The order of factors is reversed in the product inverse.

# Chapter 6

# SYSTEMS OF LINEAR EQUATIONS

## 6.1 Matrix Representation

A system of linear equations can be written in the form

$$AX = B,$$

where $A$ is the coefficient matrix, $X$ is the column of unknowns and $B$ is the column of constants. Consider

$$2x + y = 5,$$

$$x - y = 1.$$

Its matrix representation is

$$\begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}.$$

## 6.2 Inverse Matrix Method

If $A$ is invertible, then

$$AX = B.$$

Multiplying on the left by $A^{-1}$,

$$A^{-1}AX = A^{-1}B.$$

Since $A^{-1}A = I$,

$$\boxed{X = A^{-1}B}.$$

## 6.3 Gaussian Elimination

Gaussian elimination transforms an augmented matrix into row-echelon form using elementary row operations. The permitted operations are:

1. Interchanging two rows.
2. Multiplying a row by a non-zero scalar.
3. Adding a multiple of one row to another row.

### 6.3.1 Worked Example

Solve

$$x + y + z = 6,$$

$$2x - y + z = 3,$$

$$x + 2y - z = 2.$$

The augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 1 & 3 \\ 1 & 2 & -1 & 2 \end{array}\right].$$

Apply $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - R_1$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -3 & -1 & -9 \\ 0 & 1 & -2 & -4 \end{array}\right].$$

Interchange rows two and three:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -2 & -4 \\ 0 & -3 & -1 & -9 \end{array}\right].$$

Now apply $R_3 \leftarrow R_3 + 3R_2$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -2 & -4 \\ 0 & 0 & -7 & -21 \end{array}\right].$$

Back-substitution gives

$$z = 3, \qquad y = 2, \qquad x = 1.$$

Therefore,

$$\boxed{x = 1, \quad y = 2, \quad z = 3}.$$

## 6.4 Consistency of Linear Systems

A system is consistent if it has at least one solution. Otherwise, it is inconsistent.

For a system $AX = B$, the rank criterion states that the system is consistent if and only if

$$\operatorname{rank}(A) = \operatorname{rank}([A \mid B]).$$

If the common rank equals the number of unknowns, the system has a unique solution. If the common rank is smaller than the number of unknowns, a consistent system has infinitely many solutions.

# Chapter 7

# RANK OF A MATRIX

## 7.1 Definition

The rank of a matrix is the maximum number of linearly independent rows or, equivalently, linearly independent columns.

It is denoted by $\mathrm{rank}(A)$.

The rank satisfies

$$0 \leq \mathrm{rank}(A) \leq \min(m, n)$$

for an $m \times n$ matrix.

## 7.2 Rank by Row Reduction

Consider

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}.$$

Apply $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - R_1$:

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & -1 & -2 \end{bmatrix}.$$

There are two non-zero rows in echelon form. Hence,

$$\boxed{\mathrm{rank}(A) = 2}.$$

## 7.3 Rank and Linear Independence

A collection of vectors is linearly independent if the only linear combination that equals the zero vector has all coefficients equal to zero.

The columns of a matrix are linearly independent if and only if the matrix has full column rank.

For a square matrix of order $n$,

$$\mathrm{rank}(A) = n$$

if and only if $\det(A) \neq 0$.

## 7.4 Applications of Rank

Rank helps determine:

- whether a system of equations is consistent;
- whether a system has a unique solution;
- the number of independent equations;
- the dimension of the column space;
- whether the columns span the required vector space.

# Chapter 8

# EIGENVALUES AND EIGENVECTORS

## 8.1 Introduction

Eigenvalues and eigenvectors describe important properties of square matrices and linear transformations.

A non-zero vector $v$ is an eigenvector of a square matrix $A$ if

$$Av = \lambda v$$

for some scalar $\lambda$, called the eigenvalue.
Rearranging,

$$(A - \lambda I)v = 0.$$

For a non-zero solution to exist, the matrix $A - \lambda I$ must be singular. Therefore,

$$\boxed{\det(A - \lambda I) = 0}.$$

This equation is called the characteristic equation.

## 8.2 Finding Eigenvalues

Consider

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}.$$

The characteristic equation is

$$\det(A - \lambda I) = 0.$$

Thus,

$$\begin{vmatrix} 2-\lambda & 0 \\ 0 & 3-\lambda \end{vmatrix} = 0.$$

Therefore,

$$(2 - \lambda)(3 - \lambda) = 0.$$

The eigenvalues are

$$\boxed{\lambda_1 = 2, \qquad \lambda_2 = 3}.$$

## 8.3 Finding Eigenvectors

For $\lambda_1 = 2$,

$$(A - 2I)v = 0.$$

This gives

$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Hence $y = 0$, while $x$ is free. An eigenvector is

$$v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Similarly, for $\lambda_2 = 3$, an eigenvector is

$$v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

## 8.4 Important Properties

For an $n \times n$ matrix, counting eigenvalues with algebraic multiplicity over the complex numbers,

$$\mathrm{tr}(A) = \sum_{i=1}^{n} \lambda_i,$$

and

$$\det(A) = \prod_{i=1}^{n} \lambda_i.$$

Here, $\mathrm{tr}(A)$ denotes the sum of the diagonal elements.

For a real symmetric matrix, all eigenvalues are real, and eigenvectors corresponding to distinct eigenvalues are orthogonal.

## 8.5 Diagonalisation

A square matrix $A$ is diagonalizable if it can be written as

$$A = PDP^{-1},$$

where $D$ is diagonal and $P$ is invertible.

The columns of $P$ are linearly independent eigenvectors, and the corresponding eigenvalues appear on the diagonal of $D$.

Diagonalisation simplifies matrix powers:

$$A^n = PD^nP^{-1}.$$

This is useful in recurrence relations, differential equations and mathematical modelling.

# Chapter 9

# WORKED EXAMPLES

## 9.1 Addition and Subtraction of Matrices

**Example:** Find $A + B$ and $A - B$, where

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$

**Solution:**
Matrix addition is performed by adding the corresponding elements of the matrices.

$$A + B = \begin{bmatrix} 2+1 & 3+2 \\ 4+3 & 5+4 \end{bmatrix}.$$

Therefore,

$$\boxed{A + B = \begin{bmatrix} 3 & 5 \\ 7 & 9 \end{bmatrix}}$$

Similarly, matrix subtraction is performed by subtracting the corresponding elements.

$$A - B = \begin{bmatrix} 2-1 & 3-2 \\ 4-3 & 5-4 \end{bmatrix}.$$

Hence,

$$\boxed{A - B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}}$$

## 9.2 Multiplication of Matrices

**Example:** Find the product $AB$, where

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}.$$

**Solution:**
The product of two matrices is obtained by multiplying the rows of the first matrix by the columns of the second matrix.

$$AB = \begin{bmatrix} 1(2) + 2(0) & 1(1) + 2(3) \\ 3(2) + 4(0) & 3(1) + 4(3) \end{bmatrix}.$$

Simplifying, we obtain

$$AB = \begin{bmatrix} 2 & 7 \\ 6 & 15 \end{bmatrix}.$$

Therefore,

$$\boxed{AB = \begin{bmatrix} 2 & 7 \\ 6 & 15 \end{bmatrix}}$$

## 9.3 Evaluation of a Determinant

**Example:** Find the determinant of

$$A = \begin{bmatrix} 4 & 2 \\ 3 & 5 \end{bmatrix}.$$

**Solution:**
For a matrix of order two,

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

the determinant is given by

$$\det(A) = ad - bc.$$

Substituting the given values,

$$\det(A) = 4(5) - 2(3).$$

$$\det(A) = 20 - 6 = 14.$$

Hence,

$$\boxed{\det(A) = 14}$$

Since the determinant is non-zero, the matrix is invertible.

## 9.4 Finding the Inverse of a Matrix

**Example:** Find the inverse of

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}.$$

**Solution:**
The inverse of a matrix of order two is given by

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix},$$

provided that $ad - bc \neq 0$.
First, calculate the determinant:

$$\det(A) = 2(1) - 1(1) = 1.$$

Since $\det(A) = 1 \neq 0$, the inverse exists.
Using the inverse formula,

$$A^{-1} = \frac{1}{1} \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}.$$

Therefore,

$$\boxed{A^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}}$$

**Example:** Find the inverse of

For a $2 \times 2$ matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the inverse formula is:

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Find the inverse of:

$$A = \begin{pmatrix} 4 & 7 \\ 2 & 6 \end{pmatrix}$$

**Step 1: Calculate the Determinant**

$$\det(A) = (4)(6) - (7)(2) = 24 - 14 = 10$$

Since $\det(A) \neq 0$, the matrix is invertible.

**Step 2: Swap and Negate Elements**

- Swap the positions of the main diagonal elements (4 and 6).
- Change the signs of the opposite diagonal elements (7 and 2).

$$\begin{pmatrix} 6 & -7 \\ -2 & 4 \end{pmatrix}$$

**Step 3: Multiply by** $1/\det(A)$

$$A^{-1} = \frac{1}{10} \begin{pmatrix} 6 & -7 \\ -2 & 4 \end{pmatrix} = \begin{pmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{pmatrix}$$

### 9.4.1 Finding the Inverse of a $3 \times 3$ Matrix (Adjugate Method)

Find the inverse of:

$$B = \begin{pmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 4 & 1 & 8 \end{pmatrix}$$

**Step 1: Calculate the Determinant**

Expanding along the first row:

$$\det(B) = 1 \begin{vmatrix} -1 & 3 \\ 1 & 8 \end{vmatrix} - 0 \begin{vmatrix} 2 & 3 \\ 4 & 8 \end{vmatrix} + 2 \begin{vmatrix} 2 & -1 \\ 4 & 1 \end{vmatrix}$$

$$\det(B) = 1(-8 - 3) - 0 + 2(2 - (-4))$$

$$\det(B) = 1(-11) + 2(6) = -11 + 12 = 1$$

**Step 2: Find the Matrix of Cofactors**

Calculate the cofactor for each position using the sign grid $\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$:

$C_{11} = +((-1)(8) - (3)(1)) = -11$ $C_{12} = -((2)(8) - (3)(4)) = -4$ $C_{13} = +((2)(1) - (-1)(4)) = 6$
$C_{21} = -((0)(8) - (2)(1)) = 2$ $C_{22} = +((1)(8) - (2)(4)) = 0$ $C_{23} = -((1)(1) - (0)(4)) = -1$
$C_{31} = +((0)(3) - (2)(-1)) = 2$ $C_{32} = -((1)(3) - (2)(2)) = 1$ $C_{33} = +((1)(-1) - (0)(2)) = -1$

Combining these gives the cofactor matrix $C$:

$$C = \begin{pmatrix} -11 & -4 & 6 \\ 2 & 0 & -1 \\ 2 & 1 & -1 \end{pmatrix}$$

**Step 3: Transpose the Cofactor Matrix (Adjugate Matrix)**

$$\text{adj}(B) = C^T = \begin{pmatrix} -11 & 2 & 2 \\ -4 & 0 & 1 \\ 6 & -1 & -1 \end{pmatrix}$$

**Step 4: Multiply by** $1/\det(B)$

Since $\det(B) = 1$:

$$B^{-1} = \frac{1}{1}\begin{pmatrix} -11 & 2 & 2 \\ -4 & 0 & 1 \\ 6 & -1 & -1 \end{pmatrix} = \begin{pmatrix} -11 & 2 & 2 \\ -4 & 0 & 1 \\ 6 & -1 & -1 \end{pmatrix}$$

## 9.5 Solving Linear Equations Using Matrices

**Example:** Solve the following system of equations using the inverse matrix method:

$$2x + y = 7,$$

$$x - y = 2.$$

**Solution:**
The given equations can be written in matrix form as

$$AX = B,$$

where

$$A = \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}, \qquad X = \begin{bmatrix} x \\ y \end{bmatrix}, \qquad B = \begin{bmatrix} 7 \\ 2 \end{bmatrix}.$$

If $A$ is invertible, then

$$X = A^{-1}B.$$

First, calculate the determinant:

$$\det(A) = 2(-1) - 1(1) = -3.$$

Therefore,

$$A^{-1} = -\frac{1}{3}\begin{bmatrix} -1 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & -\frac{2}{3} \end{bmatrix}.$$

Now,

$$X = A^{-1}B.$$

Hence,

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & -\frac{2}{3} \end{bmatrix}\begin{bmatrix} 7 \\ 2 \end{bmatrix}.$$

Multiplying the matrices,

$$x = \frac{7}{3} + \frac{2}{3} = 3,$$

$$y = \frac{7}{3} - \frac{4}{3} = 1.$$

Therefore, the solution is

$$\boxed{x = 3, \qquad y = 1}$$

**Example: Find the inverse of** $A = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix}$

**Solution:** To find the inverse of a $2 \times 2$ matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, we use the formula:

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

**Step 1: Calculate the determinant** $\det(A)$

$$\det(A) = (3)(2) - (5)(1) = 6 - 5 = 1$$

Since $\det(A) = 1 \neq 0$, the matrix is invertible.

**Step 2: Construct the adjugate matrix**
Swap the elements on the main diagonal and change the signs of the off-diagonal elements:

$$\begin{pmatrix} 2 & -5 \\ -1 & 3 \end{pmatrix}$$

**Step 3: Multiply by** $1/\det(A)$

$$A^{-1} = \frac{1}{1} \begin{pmatrix} 2 & -5 \\ -1 & 3 \end{pmatrix} = \begin{pmatrix} 2 & -5 \\ -1 & 3 \end{pmatrix}$$

**Verification:** We can verify the result by showing that $AA^{-1} = I$:

$$AA^{-1} = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 2 & -5 \\ -1 & 3 \end{pmatrix} = \begin{pmatrix} (3)(2) + (5)(-1) & (3)(-5) + (5)(3) \\ (1)(2) + (2)(-1) & (1)(-5) + (2)(3) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

## 9.6 Finding the Rank of a Matrix

**Example:** Find the rank of the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}.$$

**Solution:**
The rank of a matrix is the maximum number of linearly independent rows or columns.
Apply the row operations

$$R_2 \to R_2 - 2R_1,$$

$$R_3 \to R_3 - R_1.$$

Then,

$$A \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & -1 & -2 \end{bmatrix}.$$

Interchanging the second and third rows gives

$$A \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix}.$$

There are two non-zero rows in the row-echelon form.
Therefore,

$$\boxed{\operatorname{rank}(A) = 2}$$

## 9.7 Finding Eigenvalues and Eigenvectors

**Example:** Find the eigenvalues and corresponding eigenvectors of

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}.$$

**Solution:**
The eigenvalues are obtained from the characteristic equation

$$\det(A - \lambda I) = 0.$$

Therefore,

$$\begin{vmatrix} 2-\lambda & 0 \\ 0 & 3-\lambda \end{vmatrix} = 0.$$

Expanding the determinant,

$$(2 - \lambda)(3 - \lambda) = 0.$$

Hence, the eigenvalues are

$$\boxed{\lambda_1 = 2, \qquad \lambda_2 = 3}$$

To find the eigenvector corresponding to $\lambda_1 = 2$, solve

$$(A - 2I)v = 0.$$

This gives $y = 0$, while $x$ can be any non-zero value. Taking $x = 1$, we obtain

$$\boxed{v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}}$$

Similarly, for $\lambda_2 = 3$, we obtain

$$\boxed{v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}}$$

Thus, the eigenvalues are 2 and 3, with the corresponding eigenvectors shown above.

# Chapter 10

# APPLICATIONS OF MATRICES

## 10.1 Applications in Economics

Matrices organise economic information such as production, cost, demand and sales.

Suppose a factory produces two products using two resources. The resource requirements per unit are

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}.$$

The first row describes the first resource and the second row describes the second resource. The columns correspond to the two products.

If production quantities are

$$x = \begin{bmatrix} 10 \\ 5 \end{bmatrix},$$

then the total resource requirements are

$$Ax = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 10 \\ 5 \end{bmatrix} = \begin{bmatrix} 35 \\ 30 \end{bmatrix}.$$

Thus, 35 units of the first resource and 30 units of the second resource are required.

## 10.2 Applications in Business

Matrices are useful for analysing sales across products and branches.

Suppose the sales matrix is

$$S = \begin{bmatrix} 20 & 30 \\ 25 & 35 \end{bmatrix},$$

where the rows represent two branches and the columns represent two products. If the unit prices are

$$p = \begin{bmatrix} 10 \\ 15 \end{bmatrix},$$

then branch revenue is calculated using

$$Sp = \begin{bmatrix} 20 & 30 \\ 25 & 35 \end{bmatrix} \begin{bmatrix} 10 \\ 15 \end{bmatrix} = \begin{bmatrix} 650 \\ 775 \end{bmatrix}.$$

The first branch earns 650 monetary units and the second earns 775 monetary units.

## 10.3 Applications in Computer Graphics

Computer graphics uses matrices to transform points, shapes and images.

A point in two dimensions can be represented as

$$v = \begin{bmatrix} x \\ y \end{bmatrix}.$$

A linear transformation is written as

$$v' = Av.$$

### 10.3.1 Scaling

The scaling matrix is

$$S = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}.$$

It transforms a point according to

$$x' = s_x x, \qquad y' = s_y y.$$

### 10.3.2 Rotation

Rotation through an angle $\theta$ about the origin is represented by

$$R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$

For $\theta = 90^\circ$,

$$R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

Applying this matrix to the point $(2, 1)$,

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

The point is rotated $90^\circ$ counterclockwise.

### 10.3.3 Reflection

Reflection in the $x$-axis is represented by

$$F_x = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Thus,

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ -y \end{bmatrix}.$$

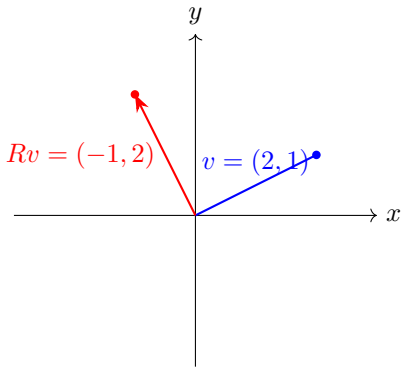



Figure 10.1: A point transformed by a rotation matrix.

## 10.4 Applications in Network Analysis

Networks can represent roads, communication systems, friendship connections and computer networks.

An adjacency matrix records which vertices are connected. For a simple undirected graph, the adjacency matrix is symmetric.

Consider a network with three vertices, where vertex 1 is connected to vertices 2 and 3, while vertices 2 and 3 are not connected to each other.

Its adjacency matrix is

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

The entry $a_{ij} = 1$ indicates a connection from vertex $i$ to vertex $j$; zero indicates no direct connection.

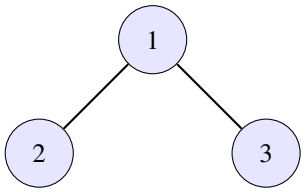



Figure 10.2: A simple network represented by an adjacency matrix.

The entry $(A^2)_{ij}$ counts the number of walks of length two from vertex $i$ to vertex $j$, for this unweighted adjacency-matrix representation.

## 10.5 Applications in Statistics and Data Analysis

A dataset with several observations and variables can be arranged in a matrix.

Suppose three students have marks in two subjects:

$$X = \begin{bmatrix} 70 & 80 \\ 60 & 75 \\ 90 & 85 \end{bmatrix}.$$

Rows represent students and columns represent subjects. This arrangement simplifies calculations of totals, averages and other statistical quantities.

In multivariate statistics, covariance matrices describe relationships among variables. If $X$ is a centred data matrix with $n$ observations and $p$ variables, one sample covariance matrix is

$$S = \frac{1}{n-1} X^T X, \qquad n > 1.$$

The resulting $p \times p$ matrix contains variances on the diagonal and covariances off the diagonal.

## 10.6 Applications in Engineering

Engineers use matrices to model interconnected systems, electrical networks, structural forces and control systems.

For example, a system of circuit equations may be represented by

$$RI = V,$$

where $R$ is a resistance coefficient matrix, $I$ is a vector of unknown currents and $V$ is a voltage vector. Depending on the circuit model, these quantities and matrices can be defined to represent the relevant equations.

Solving the matrix system helps determine the unknown currents when the system has a unique solution.

## 10.7 Applications in Physics

Matrices represent coordinate transformations, rotations, coupled physical systems and certain quantum-mechanical operators.

A linear system of differential equations can be written as

$$\frac{dX}{dt} = AX.$$

When the matrix $A$ has a suitable eigenvector decomposition, eigenvalues and eigenvectors can help describe the behaviour of the system over time.

## 10.8 Applications in Computer Science

Matrices are used in image processing, machine learning, search algorithms, data representation and numerical computation.

An image can be stored as a matrix of pixel intensities. A colour image is often represented using separate matrices or channels for its colour components.

In machine learning, matrices represent collections of data and model parameters. Matrix multiplication is a central operation in many neural-network computations.

## 10.9 Applications in Population Models

A simple population model can use a transition matrix to represent how a population changes between categories.

If $x_t$ is the population vector at time $t$, then

$$x_{t+1} = Ax_t.$$

Repeated application gives

$$x_t = A^t x_0.$$

# CONCLUSION

Matrices provide an organised and powerful method for representing numerical information and solving mathematical problems.

This project examined the fundamental definitions, classification and algebraic operations of matrices. It also discussed determinants, inverse matrices, rank, systems of linear equations, eigenvalues and eigenvectors.

Worked examples demonstrated the calculation of matrix products, determinants, inverses and solutions of linear systems. The mathematical results showed how these concepts are connected and why the conditions for invertibility and consistency are important.

The applications presented in the project demonstrate the importance of matrices in economics, business, computer graphics, network analysis, engineering, statistics, physics and computer science.

In conclusion, matrix theory is an essential part of modern mathematics. A clear understanding of matrices provides a foundation for advanced studies in linear algebra, numerical analysis, mathematical modelling and computational mathematics.

# Chapter 11

# REFERENCES


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  *Description:* The definitive reference for graduate-level matrix theory. It covers eigenvalues, canonical forms, matrix factorizations, and special classes of matrices (Hermitian, positive definite).
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  *Description:* A classic treatise providing a deep algebraic perspective on matrix polynomials, functions of matrices, and stability theory.
- **Advanced Linear Algebra** – Steven Roman
  *Description:* A thorough, graduate-level approach to modules, vector spaces, inner product spaces, and structure theorems.
- **Matrix Analysis and Applied Linear Algebra** – Carl D. Meyer
  *Description:* Excellent transition textbook from basic computations to practical applications, detailing Markov chains, animations, and page-ranking systems.
- **Matrix Methods in Data Mining and Pattern Recognition** – Lars Eldén
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